

$$\Delta N_{\text{eff}} < 0.281$$

$$\Delta N_{\text{eff}} < 0.0738$$

$$\Delta N_{\text{eff}} = 0.108^{+0.021}_{-0.031}$$

- *Planck* PR3
- S4-like (Λ CDM Fid.)
- S4-like (LiMR Fid.)

$$\log_{10}(z_{\text{NR}}) < 3.49$$

$$\log_{10}(z_{\text{NR}}) < 3.16$$

$$\log_{10}(z_{\text{NR}}) = 3.966^{+0.096}_{-0.13}$$

$$\sigma_{\text{LN}} = 0.58^{+0.26}_{-0.38}$$

